

BIOINFORMATICS

WEEK 1 – DAY 7

FROM DNA CODE TO 3D SHAPE

Your body uses a precise blueprint system to build the molecular machines that keep you alive:

- **The Blueprint (DNA):** The process starts with a genetic code, like ATG.
- **The Copy (RNA):** The cell transcribes this DNA code into a temporary RNA message.
- **The Assembly (Protein):** The RNA is translated into a long, linear chain of amino acids.

However, a linear chain cannot do any work. To become functional, the protein must fold into a highly specific three-dimensional shape.

WHY SHAPE IS EVERYTHING

In biology, shape determines function. The specific 3D structure of a protein controls four critical outcomes -

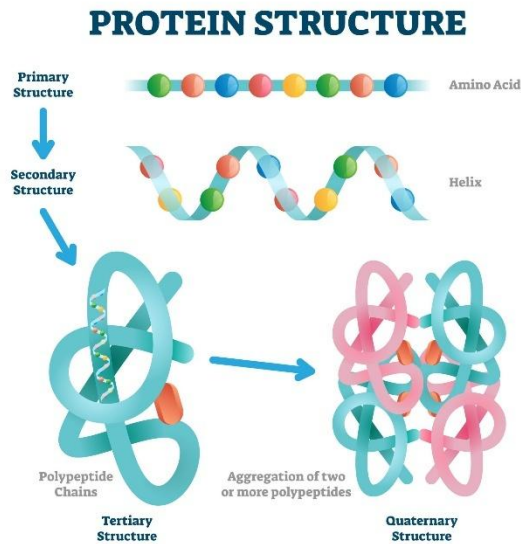
- **Molecular Binding:** It dictates exactly which molecules (like nutrients, oxygen, or drugs) the protein can latch onto, acting like a lock and key.
- **Biological Function:** It determines the protein's role, such as acting as a structural building block or an enzyme that speeds up chemical reactions.
- **Correct Operation:** If the protein folds into the exact right shape, it works perfectly. If it misfolds, it fails.
- **Mutation Impact:** Looking at the 3D shape helps scientists predict whether a genetic mistake will be harmless or cause a severe disease.

THE POWER OF A SINGLE CHANGE

Because a protein's shape relies on the perfect sequence of amino acids, changing just one single link in that chain can cause the entire structure to fail. A classic example of this is Sickle Cell Disease:

1. A genetic mutation causes one single amino acid to be substituted for another in hemoglobin (the oxygen-carrying protein in blood).
2. This tiny change alters the protein's surface, causing hemoglobin molecules to stick together abnormally.
3. These rigid protein clumps force the entire red blood cell into a rigid, crescent "sickle" shape, blocking blood flow and causing pain.

The structure of a protein is divided into four distinct levels - primary, secondary, tertiary, and quaternary. which determine its unique 3D shape and biological function. Proteins are built from chains of **amino acids** folded into complex, specific configurations.



THE FOUR LEVELS OF PROTEIN STRUCTURE

- **Primary Structure:** The linear sequence of amino acids linked by peptide bonds.
- **Secondary Structure:** Local folding into alpha-helices or beta-pleated sheets stabilized by hydrogen bonds.
- **Tertiary Structure:** The overall three-dimensional shape of a single polypeptide chain, driven by hydrophobic interactions and disulfide bridges.
- **Quaternary Structure:** The arrangement and alignment of multiple polypeptide chains (subunits) working together as a single functional unit.

Detailed Structural Comparison

Structural Level	Main Bonding Types	Common Visual Shapes
Primary	Covalent peptide bonds	Linear bead-like chain
Secondary	Hydrogen bonds	Coiled helix or flat pleated sheets
Tertiary	Hydrophobic interactions, ionic, disulfide bonds	Complex globule or fibrous fold
Quaternary	Hydrophobic and ionic bonds between subunits	Multi-subunit protein complex

KEY CHEMICAL BONDS HOLDING PROTEINS TOGETHER

- **Peptide Bonds:** Strong covalent links between the amino group of one amino acid and the carboxyl group of another.
- **Hydrogen Bonds:** Weak bonds between oxygen and hydrogen atoms that stabilize secondary shapes.
- **Disulfide Bridges:** Strong covalent bonds formed between the sulfur atoms of two cysteine amino acids.
- **Hydrophobic Interactions:** The crowding of non-polar amino acid side chains toward the interior of the protein to escape water.

THE PROTEIN DATA BANK (PDB)

the world's single, global repository for the three-dimensional structural data of biological macromolecules, such as proteins, DNA, and RNA. Established in 1971, it acts as a massive, free digital library that allows scientists to download and visualize the exact 3D coordinates of molecules.

How PDB Works

- **Data Submission:** Researchers worldwide determine protein structures using experimental methods and upload the 3D atomic coordinates to the PDB.
- **Curated Archives:** Before public release, expert biocurators review and validate the data to ensure accuracy and standardized formatting.
- **Open Access:** Anyone—from high school students to Nobel laureates—can access, search, and download these structural files completely free of charge.
- **Unique ID:** Every single structure deposited in the database is assigned a unique, four-character alphanumeric identifier known as a **PDB ID** (e.g., 1A3N for human hemoglobin).

The Primary Methods Used to Find PDB Structures

- **X-ray Crystallography:** The most common method. Scientists grow a crystal of the protein and blast it with X-rays to map the electron density of its atoms.
- **Nuclear Magnetic Resonance (NMR) Spectroscopy:** Used for smaller proteins. It uses strong magnetic fields to map atomic interactions while the protein is naturally dissolved in a liquid solution.
- **Cryo-Electron Microscopy (Cryo-EM):** The fastest-growing method. It flash-freezes proteins in a thin layer of ice and uses a high-powered electron microscope to snap thousands of individual images to build a 3D model.

Key Information Found in a PDB Entry

- **Protein Name:** The official biological name of the molecule (e.g., Insulin).
- **Organism:** The specific source species from which the protein was obtained (e.g., *Homo sapiens*).

- **Authors:** The names of the scientists who successfully isolated, solved, and deposited the structure.
- **Experimental Method:** The technique used to determine the structure (e.g., X-ray Crystallography or Cryo-EM).
- **Resolution:** A metric measured in Angstroms (Å) for crystallography; lower numbers mean higher, sharper atomic detail.
- **Chains:** The distinct individual polypeptide strands that make up the complete protein (e.g., Chain A and Chain B).
- **Ligands:** Non-protein molecules, ions, or drugs bound to the protein structure (e.g., Zinc ions in insulin).
- **Sequence:** The exact, ordered chain of amino acids listed in a standard single-letter or three-letter text code.
- **3D Viewer:** An interactive online tool (like Mol*) that lets you rotate, zoom, and change styles (like ribbon or space-filling) of the molecule.

PyMOL

- a powerful, open-source molecular visualization system used by scientists to create high-quality 3D images and animations of proteins, nucleic acids, and other biomolecules.
- It reads the raw 3D atomic coordinates from PDB files and renders them into clear, customizable visual models.

Why Scientists Use PyMOL -

- **Publication Graphics:** It is the industry standard for creating the 3D protein diagrams seen in scientific journals and textbooks.
- **Structural Analysis:** Researchers use it to measure exact distances between atoms, calculate hydrogen bonds, and inspect active binding sites.
- **Drug Design:** Chemists use PyMOL to visually inspect how well a synthetic drug molecule docks inside a target viral or bacterial protein.

Opening Human Insulin

- Open the insulin structure in PyMOL.
- You should see a rotating 3D model.
- Spend a few minutes simply rotating and zooming the molecule.
- Understanding the shape is the first step.

Basic PyMOL Skills

1. **Rotate** : Click and drag.
2. **Zoom** : Use the mouse wheel.
3. **Move** : Right-click and drag (or use the appropriate mouse mode depending on your setup).
4. **Display Styles** : Try different representations: Cartoon, Sticks, Spheres, Surface. Each highlights different structural features.

WHAT ARE PROTEIN CHAINS?

Many complex proteins are not just one single continuous loop. Instead, they are made of multiple individual amino acid strands that bind together to form a functional machine. In biology, each of these independent strands is called a protein chain.

The Structure of Insulin

Insulin is a classic example of a multi-chain protein. It consists of exactly two distinct chains working together:

- **Chain A**: A short peptide strand containing 21 amino acids. It features an internal disulfide loop that helps lock its shape.
- **Chain B**: A slightly longer peptide strand containing 30 amino acids.

How They Connect

- **Disulfide Bridges**: Chain A and Chain B are held tightly together by two strong, covalent chemical bonds called disulfide bridges (sulfur-to-sulfur connections).
- **The Result**: Together, these 51 amino acids form the complete, active insulin hormone that your body uses to regulate blood sugar levels.

WHAT IS AN AMINO ACID RESIDUE?

When single amino acids link together to form a protein chain, they undergo a chemical reaction that releases a molecule of water. Because a small part of the original amino acid is lost in this process, what remains in the chain is technically called an amino acid residue.

Position Numbers: The Protein's Map

To navigate a protein, scientists assign a unique position number to every single residue in the chain, starting from number 1 at the beginning (the N-terminus) and counting up sequentially to the end. This acts like a street address, allowing researchers to pinpoint exactly where an amino acid is located.

Examples from Insulin

Your three examples represent specific "addresses" found within the chains of insulin:

- **Glycine (Gly):** Found at **Position 1 on Chain A**. It is the very first amino acid residue that starts the A chain.
- **Serine (Ser):** Found at **Position 9 on Chain A**. It sits right in the middle of a flexible loop region.
- **Cysteine (Cys):** Found at multiple critical locations, such as **Position 7 on Chain A** and **Position 7 on Chain B**. These specific addresses are crucial because the cysteines reach out across the space to form the disulfide bonds that lock Chain A and Chain B together.

THE DIFFERENT WAYS TO COLOR A PROTEIN

You can change how PyMOL colors a molecule depending on what structural features you want to highlight:

- **Color by Chain:** Assigns a distinct color to each separate polypeptide strand (e.g., **Chain A -> Green, Chain B -> Cyan**). This makes it easy to see where one independent chain ends and the next begins.
- **Color by Secondary Structure:** Colors the protein based on its local folds. Typically, alpha-helices turn **red/magenta**, beta-sheets turn **yellow**, and flexible loops turn **green**. This instantly shows you the structural scaffolding of the protein.
- **Color by Element:** Colors every individual atom by its standard chemical identity (e.g., **Carbon -> Green/Gray, Oxygen -> Red, Nitrogen -> Blue, Sulfur -> Yellow**). This is crucial when zooming in to see exactly how chemical bonds form.

How to Do It in PyMOL

1. Look at the Object Menu Panel on the right side of the screen.
2. Find your protein's name and click the **[C]** button (which stands for Color).
3. Select from the pop-up menu: by chain, by ss (secondary structure), or by element.

THE RELATIONSHIP BETWEEN SEQUENCE AND STRUCTURE

Biologists use a foundational rule: the linear sequence of amino acids dictates how a protein folds into its 3D structure, which in turn determines its function. However, comparing 3D shapes tells a much deeper story than just reading the text code.

The General Rule: Similar Sequence = Similar Structure

- **Evolutionary Conservation:** If two proteins share a high percentage of identical amino acids (high sequence similarity), they usually fold into almost identical 3D shapes.
- **Predictable Folding:** This predictable pattern is how modern AI tools and software can guess a new protein's shape by comparing it to known sequences in databases like the PDB.

The Exceptions: When the Rule Fails

While a similar sequence *often* means a similar structure, there are critical exceptions where this assumption breaks down:

- **The Tipping Point:** Sometimes, a sequence change of just one or two critical amino acids occurs at a vital hinge or core stability point. This tiny change can completely disrupt the delicate balance of forces, causing the protein to misfold or collapse.
- **Metamorphic Proteins:** Some rare proteins have a single amino acid sequence that can switch back and forth between two completely different 3D shapes depending on its environment.
- **Engineered Dual-Shapes:** Scientists have successfully designed artificial protein sequences that are 98% identical but fold into completely unique, unrelated 3D architectures.